COMP 3920

Database Systems 1

Practical Midterm

Time Permitted: 2 Hours (for both the Written and Practical Portions)

TOTAL Score
/ 60

INSTRUCTIONS:

This exam is **Open Book**.

What is Allowed

- ☑ You may consult any of your class notes and/or previous labs/projects.
- ☑ You may consult the Internet.
- ☑ You may consult your textbook.

What is NOT Allowed

- ☒ You may NOT share any portion of your solutions with others.
- ☒ You may NOT use anyone else's solutions as your own.
- ☒ You may NOT have someone else complete any portion of your exam.
- ☒ You may NOT discuss your solution with others.
- ☒ You may NOT work with anyone else to complete the exam.

This exam consists of the following sections to complete:

Written:

SECTION 1:	SELECT Queries	20 Marks

Practical:

SECTION 2:	Bad Design to Good Design	10 Marks
SECTION 3:	Fix the Node Code	15 Marks
SECTION 4:	Node Navigation Breadcrumbs	15 Marks

Before starting the Exam:

Go to the following website:

http://onlinequiz.click

Login using your BCIT Student number and password shown below:

Student ID: **<Your BCIT Student Number>**
Password: **Perceptive&Practical**

This Online Quiz Site (http://onlinequiz.click) will provide instructions you will need to complete this midterm.

Hi, Patrick Guichon!

Instructions

Steps:

1. Download the code to fix here:

code.zip

You will need this code for the section: "Fix the Node Code".

SECTION 2: Bad Design to Good Design

SECTION 2
Score:
/ 10

The following table is an example of a bad design.
Redesign the table to reduce redundancy and provide a consistent data structure.

Project

project_id	project_name	budget_cost	budget_hours	completion_date	employees
1	install left turn signal at 104th Av	$100,000	25	May 5, 2021	**Breana Haroldson***, Ben Meier, Harish Loman
2	Broadway Repaving	$1,200,000	90	Feb 20, 2019	**Edward Ng***, Harish Loman, Mark Allegri, Puabi Dahl, Usagi Yago
3	Replace Median on Main St	$550,000	75	NULL	**Edward Ng***, Ben Meier, Breana Haroldson, Usagi Yago
4	Street Line Repainting on 32nd Ave	$350,000	25	NULL	**Ben Meier***, Usagi Yago
5	Street Lamp Upgrades on east-side	$450,000	85	May 24, 2023	**Ben Meier***, Mark Allegri, Puabi Dahl

Notes:

* indicates the **site supervisor** for the project. (There will be only 1 site supervisor per project.)
All columns are required, except `completion_date` which may contain NULL.

Step 1: Redesign this table and represent your good design with an ERD.

Using Draw.io, redesign the above table using good database design techniques.
Draw an ERD and all the applicable relationships to represent your normalized database.

Add the appropriate surrogate keys, primary and foreign keys, relationship lines and cardinality.

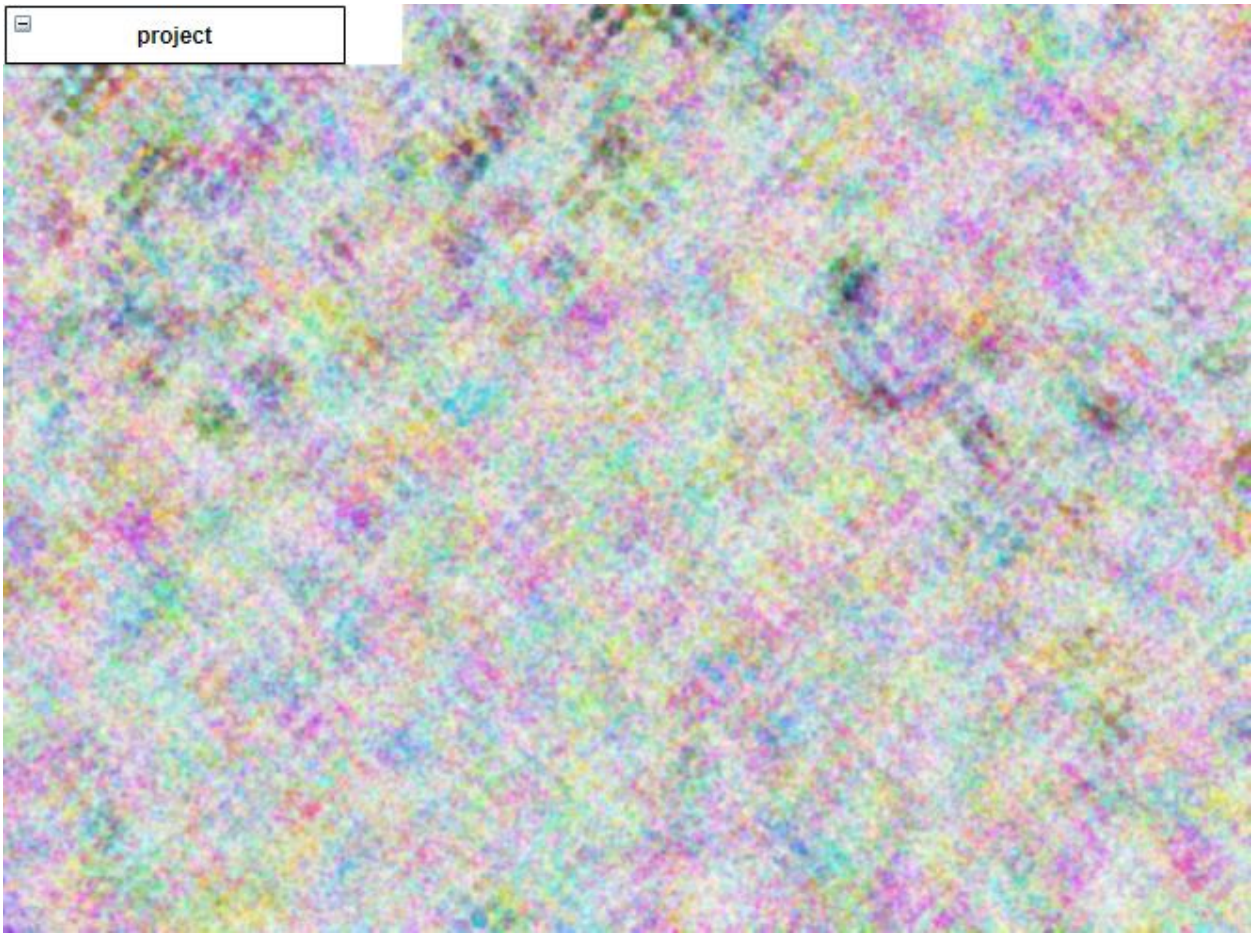
Make sure that relationship lines link to the correct attributes on the diagram.

You do NOT need to specify the data types for each attribute.

Submit your ERD.

Export your Draw.io ERD as a PNG file.
Filename: **01_Project_Good_Design_ERD.png**

Submit



SECTION 3: Fix the Node Code

SECTION 3
Score:
/ 15

A junior web programmer attempted to create a simple website using Node.js. They've made a few mistakes and as a result the code doesn't work properly.

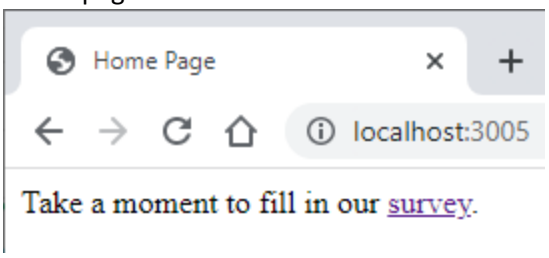
It's your job to fix the code and complete the website.

There are **5 mistakes** in the website.

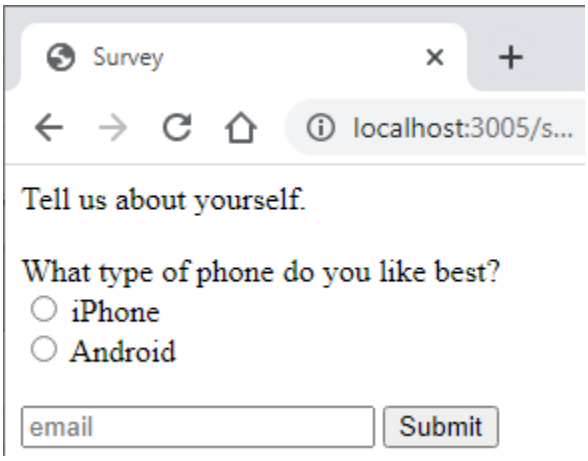
Find them and fix them.

The website was supposed to have the following pages.

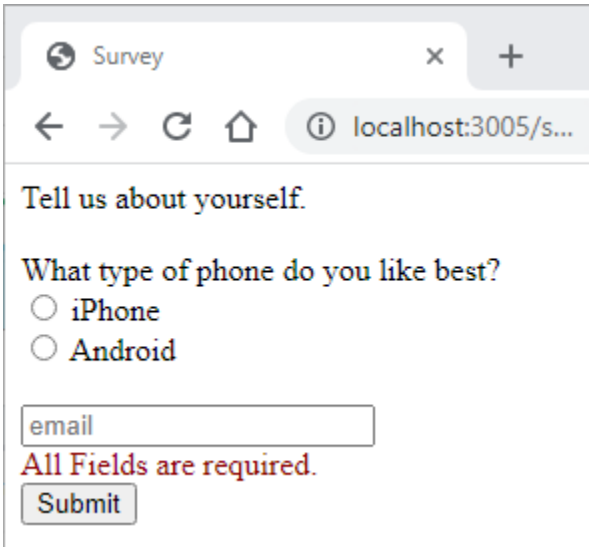
Home page:



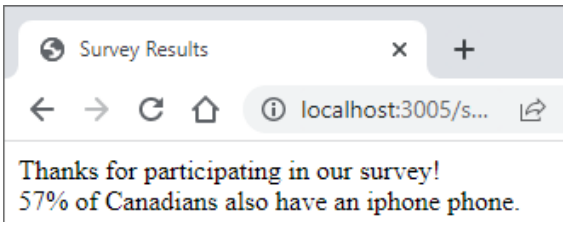
Clicking on the survey link should take you here:



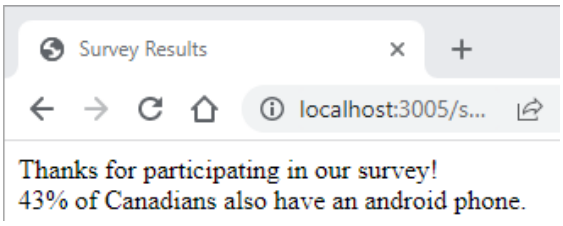
If you don't select one of the phone types, or you don't put in your email address, it should respond with: All Fields are required.



If you properly enter a phone type and your email address, it should respond by telling you how many other people in Canada also share the same phone type as you (stats as of 2022):



OR



(if you chose *iphone*) (if you chose *android*)

Step 1: Download the starting code from the link provided at https://onlinequiz.click.

2. Download the code to fix here:

code.zip

You will need this code for the section: "Fix the Node Code".

Step 2: Modify the code to fix the errors.

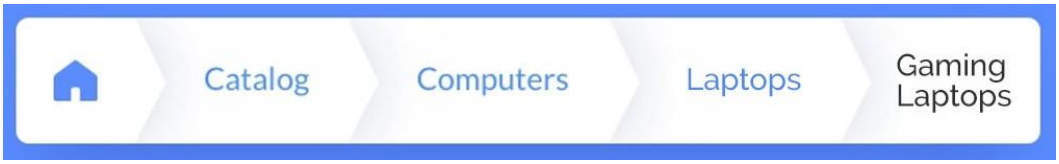
Note: You'll submit your code for this section as part of the submission for the next section.
(There is **no** need to submit the code once for this section and again for the next section. Submit the code after completing both sections.)

SECTION 4: Node Navigation Breadcrumbs

SECTION 4
Score:
/ 15

Using the node code from the previous section you will add new routes and new ejs views in order to add the following functionality.

The following is an example of a navigation breadcrumb:



https://media.licdn.com/dms/image/D4D12AQGgAkiz69zxKQ/article-cover_image-shrink_720_1280/0/1686823116182?e=2147483647&v=beta&t=PVk4cXE7ZKinr53hGS2IzLWbr8TZTfoOiFUXbLY67eo

This shows that the user is on the "Gaming Laptops" page. The "Gaming Laptops" page is a subpage under "Laptops", which is under "Computers" and under "Catalog". These links help see where you are on the site and help you go to a different section of the site. For example, you could click on "Laptops" to see other categories of laptops other than gaming laptops.

Your task is to create a simple version of these navigation breadcrumbs and a few views to demonstrate them.

IMPORTANT: Create a **common header** that can be used in all pages so that you don't copy/paste the navigation breadcrumbs in each page.

HINT: You can use the following code to get an array of the path folders:

```
let fullUrl = req.protocol + '://' + req.get('host') + req.originalUrl;
folders = new URL(fullUrl).pathname.split("/").slice(1);
console.log(folders);
```

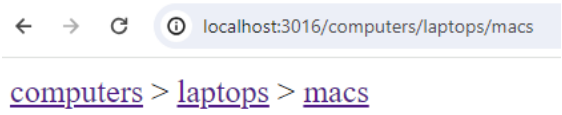
Produces the following output:

```
[ 'computers', 'laptops', 'macs' ]
```

When going to the URL: `/computers/laptops/macs`

Your pages should look like this:

Example 1 - URL: `/computers/laptops/macs`

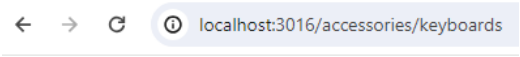


computers > laptops > macs

macs

(clicking on "computers" links to `/computers`, clicking on "laptops" links to `/computers/laptops` and clicking on "macs" links to `/computers/laptops/macs`)

Example 2 - URL: `/accessories/keyboards`

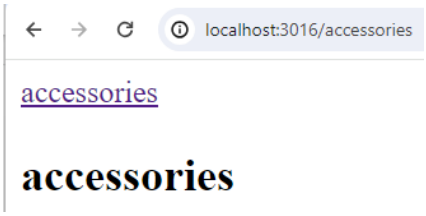


accessories > keyboards

keyboards

(clicking on "accessories" links to `/accessories` and clicking on "keyboards" links to `/accessories/keyboards`)

Example 3 - URL: `/accessories`



accessories

accessories

(clicking on "accessories" links to `/accessories`)

Your code should have routes for the following pages:

`/accessories`

`/accessories/keyboards`

`/accessories/mice`

`/computers/`

`/computers/laptops`

`/computers/laptops/macs`

`/computers/laptops/windows`

`/computers/desktops`

`/computers/desktops/macs`

`/computers/desktops/pcs`

Submit your final code with the errors fixed (from previous section) <u>and</u> the new functionality (this section).
Zip the code into a single zip file.
Do **NOT** include the `/node_modules/` folder in your zip file.

Submit

02_Node_Code.zip

Congratulations! You are done!